QP CODE: 24900027

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi - Disciplinary Course - MGIMDCBIF100 - BIOINFORMATICS FOR BEGINNERS

(2024 ADMISSION ONWARDS**)**

Duration: 1.5 Hours Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

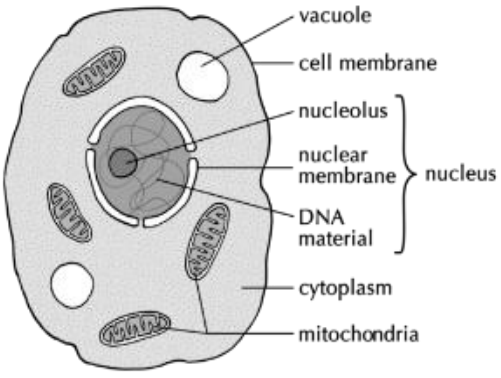
Students should attend at least one question from each course 0utcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Short Answer Questions
Answer any 5 out of 7 questions
Each questions carriers 2 marks

1. Outline basic concepts in a cell. [K] [1,2]
   Cells are the structural, functional, and biological units of all living beings. Replicate itself independently. Known as the building blocks of life. Each cell contains a fluid called the cytoplasm, which is enclosed by a membrane.
2. Demonstrate the structure of a cell [U] [1,2]



3. Outline the structure of protein. [K] [1,2]
   Proteins are polymers made up of amino acids, which are joined together by peptide bonds. The shape of a protein is important to its function, as proteins act as building blocks for cells and enzymes. The structure include Primary structure, Secondary structure, Tertiary structure, Quaternary structure.
4. Identify the significance of internet in bioinformatics. [K][3]

Provides a platform for accessing and sharing biological data, and for collaborating globally to Access data, Tools, Educational resources, scientific information, Access for scientists etc.

5. Write on the chromosome which is completely sequenced first. [U][4]
   Chromosome 22 is the first of 23 human chromosome pairs to be deciphered because of its relatively small size and its association with several diseases and because of the groundwork of several scientists beginning in the early 1990s.
6. What is the main importance of GenBank database? [K][5,6]
   It a comprehensive database that contains publicly available nucleotide sequences for more than 300 000 organisms named at the genus level or lower, obtained primarily through submissions from individual laboratories and batch submissions from large-scale sequencing projects.
7. Define sequence identity. [K][5,6]
   Sequence identity refers to the degree of similarity between the DNA or protein sequence with another sequence or database sequences. It's crucial for functional annotation, structure prediction, establishing evolutionary relationships, disease diagnosis etc.

(5*2=10)

Part B

Short Essay Type Questions
Answer any 3 out of 5 questions
Each questions carriers 5 marks

8. Differentiate between prokaryotic and eukaryotic cell [U][1,2]
   Introduction- cell, parts and type (2 marks): A cell is the smallest unit of life, and is made up of three parts: the membrane, the nucleus, and the cytoplasm. Two types include prokaryotic and eukaryotic cells.
   Difference (3 marks).

Prokaryotic Cells	Eukaryotic Cells
Very minute in size	Fairly large in size
Nuclear region (nucleoid) not surrounded by a nuclear membrane	Nuclear material surrounded by a nuclear membrane
Single chromosome present	More than one chromosome present
Nucleolus absent	Nucleolus present
Membrane bound cell organelles are absent	Membrane bound cell organelles are present
Cell division by fission or budding (no mitosis)	Cell division by mitosis or meiosis

9. Classify different levels of hierarchical organization. [U][1,2]
   Explaining Hierarchical organization at atomic, molecular, cellular, tissue, organ, organism, population, community, ecosystem, and biosphere levels (5 mark)
10. Differentiate between global and local alignment. [U][5,6]
    Introduction (2 marks): sequence alignment is a way of arranging the sequences of DNA, RNA, or protein to identify regions of similarity. Global alignment**:** Aligns the entire length of both sequences, usually done for sequences of high similarity. Local alignment: Done to identify local regions of similarity/ conserved regions.
    Difference (3 marks): Global alignment: Tries to align entire sequence, Align all letters from query and target, Suitable for closely related sequences. A general Global Alignment Method is NEEDLEMAN - WUNSCH. Local alignment: Align regions having highest

similarities, Align substring of target with substring of query, Suitable for more divergent sequences, a general Local Alignment Method is SMITH-WATERMAN

11. Illustrate the process of retrieving data from UniProt? [U][5,6]
    Introduction to UniProt (2 marks): An annotated database of protein sequences and functional information.
    Process (3 marks): Go to the UniProt website , Search for the protein by accession or perform a keyword query, select entry by protein name/ organism, select the sequence tab, Copy and paste the sequence, save the sequence as a file.
12. Summarize history, aim and scope of Bioinformatics. [U][3]
    History (1 mark): Ben Hesper and Paulien Hogeweg, Margaret Dayhoff and her collaborators.
    Aim (2 mark): Storage of biological data in form of databases, Development of tools and resources for data analysis, Exploitation of computational tools to analyse the biological data and interpret the analysed data in a biologically meaningful manner.
    Scope 2 mark
    Mention any three of; Management and analysis of a wide set of biological data, genomic basis of traits and diseases, research and development of the biomedical field, establishing evolutionary relationships, predicting structures of proteins, crop improvement endeavours, reverse vaccinology. Personalised medicine etc.

Part C

Long Essay Type Questions
Answer any 1 out of 3 questions
Each questions carriers 10 marks

13. What are the implication of HGP for future medical research and treatment? [U][4]
    Introduction to HGP (start date, countries contributed to the project, aim, outcomes (2 marks)
    Explaining applications of HGP like Identifying disease-linked variants, disease susceptibility prediction, screening, Matching patients to the best treatment/ personalised therapies, Designing more effective drugs, Predicting drug responsiveness, toxicity prediction, in silico vaccine design, cancer genomics, clinical genomics. (8 marks).
14. Explain hierarchical organization and classify each level. [U][2]
    Explanation for each level of Hierarchical organization beginning with atomic, molecular, cellular, tissue, organ, organism, population, community, ecosystem, biosphere. (10 marks)
15. Describe the difference between FASTA and BLAST in terms of Functionality. [U][5,6]
    Defining BLAST and FASTA (2 marks)
    Differentiation of functionality (8 mark):

BLAST	FASTA
BLAST stands for Basic Local Alignment Search Tool	FASTA “fast-all”.
Faster	slower
Return multiple high scoring alignments	Provide only one high scoring alignment
Uses substitution matrix to find matching word	Use hashing technique to find matching ktups
Better for similarity search in homologous sequence	Similarity searches in unrelated sequence
Better for protein searches	Better for nucleotide searches
Gaps are allowed	Gaps are not allowed
BLAST searches similarities in local alignment by comparing individual residues in the two sequences	FASTA searches similarities in local alignments by comparing sequence patterns or words
High specificity than fasta	Low specificity

END OF THE QUESTION PAPER

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